Recurrence Relation → To get the next term using previous k terms.

Linear → no powers.

$a_n = 2a_{n-1} - 3a_{n-1}$ ✓ $a_n = (a_{n-1})^2$ not linear

Homogenous → nothing extra added

e.g $a_n = 2a_{n-1} - 3a_{n-1}$

$a_n = 2a_{n-1} + 5$
not homogenous.

current term you want to find — constants — 1 step back term → recurrence relation

$a_n = c_1 a_{n-1} + c_2 a_{n-2} \cdots$

Degree 1 uses $c_1 a_{n-1}$

Degree 2 uses $c_2 a_{n-2}$

⋮

$a_n = 3a_{n-1} - a_{n-2}$
→ degree 2

$a_n = 2a_{n-1} - 5a_{n-3}$
→ degree 3

all
this are
recurrence
Relation

$a_n = a_{n-5}$ degree 5

Solving Linear recurrence Relations with constant coefficients.

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

$a_n = r^n$ → Assume Solution

Plug in the equation

$$r^n = c_1 r^{n-1} + c_2 a r^{n-2}$$

and then divide by r^{n-k} → where k is degree

to get this form e.g

$r^2 - r - 2 = 0$ → characteristic eqn

find the roots r_1 and r_2 and this only happens if degree = 2

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

↓

general sol

lets say if roots are repeated ($r_1 = r_2 = r_3$)

Use this formula

$$a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$$

e.g $r = 3$

$$a_n = \alpha_1 (3)^n + \alpha_2 n (3)^n$$

multiplicity of a root → how many times the root repeats itself.

If we get roots like this

$r = 2, 2, 5, 5, 7$

$2, 2 \rightarrow \alpha_{1,0}, \alpha_{1,1}$

$5, 5 \rightarrow \alpha_{2,0}, \alpha_{2,1}$

$7 \rightarrow \alpha_{3,0}$

To write multiple roots in a general sol

general sol

$$a_n = (\alpha_{1,0} + \alpha_{1,1} n + \alpha_{1,2} n^2) 2^n \ldots$$

$$a_n = (\alpha_{1,0} + \alpha_{1,1} n) 2^n + (\alpha_{2,0} + \alpha_{2,1} n) 5^n + (\alpha_{3,0}) 7^n$$

total degree = $2+2+1 \rightarrow 5$

$2+2+1$: $\downarrow$ no. of 2's, $\downarrow$ no. of 5's

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find sol to the recurrence relation:

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

$a_0 = -1$

$a_1 = -2$

$a_2 = -1$

$a_n = r^n$

$$r^n = -3r^{n-1} - 3r^{n-2} - r^{n-3} \quad / \; r^{n-3}$$

$$r^3 = -3r^2 - 3r - 1$$

$$r^3 + 3r^2 + 3r + 1 = 0$$

$$(r+1)^3 = 0 \qquad r = -1 \qquad r_1 = -1 \quad r_2 = -1$$

general sol

$$a_n = (\alpha_{1,0} + \alpha_{1,1} n + \alpha_{1,2} n^2)(-1)^n$$

$$a_n = (\alpha_1 + \alpha_2 n + \alpha_3 n^2)(-1)^n$$

$$a_n = (\alpha_1)(-1)^n + (\alpha_2 n)(-1)^n + (\alpha_3 n^2)(-1)^n$$

using intial conditions to solve

$a_0 = -1$
$a_1 = -2$
$a_2 = -1$

$$a_0 = 1 = (\alpha_1)(-1)^0 + (\alpha_2 0)(-1)^0 + (\alpha_3 (0)^2)(-1)^0$$

$$\boxed{1 = \alpha_1}$$

for $a_1 = -2$ it will give eq $\boxed{\alpha_2 + \alpha_3 = -1}$

for $a_2 = -1$ it will give eq $\boxed{\alpha_2 + 2\alpha_3 = -1}$

after finding all alphas plug in back

$$a_n = (\alpha_1)(-1)^n + (\alpha_2 n)(-1)^n + (\alpha_3 n^2)(-1)^n$$

$$a_n = (-1)^n + (-1n)(-1)^n$$

$$= (-1)^n - (-1)^n n$$

$$= \boxed{(-1)^n (1 - n)}$$

finding non-homogenous linear recurrence relations.

$$a_n = 3a_{n-1} + \boxed{2n} \qquad \text{with } a_1 = 3$$

non homogenous as we have a non-a term.

To solve it consider the homogenous part.

$$\boxed{a_n = 3a_{n-1}}$$ first

$a_n = r^n$

$r^n = 3r^{n-1}$ divide by r^{n-1}

$$\boxed{r = 3}$$

ie $a_n = d_1(r^n) = \boxed{d_1(3^n)}$

Now finding for particular sol trial is $2n$.

the form is $a_n = cn + D$

$$\boxed{a_n = 3a_{n-1} + 2n}$$

↙ Substitute the recurrence

$$3[c(n-1) + D] + 2n = cn + D$$

$$3(Cn - 3C + 3D + 2n = Cn + D$$

$$3(Cn + 2n - 3C + 3D = Cn + D$$

$$n(3C + 2) - 3C + 3D = Cn + D$$

$C = 3C + 2$

$C - 3C = 2$

$-2C = 2$

$\boxed{C = -1}$

$D = 3D - 3C$

$D - 3D = -3C$

$-2D = +3C$

$2D = 3C$

$2D = -3$

$\boxed{D = -\frac{3}{2}}$

Particular Sol

$Cn + D$

$$\boxed{-1n - \frac{3}{2}}$$

now with the general sol

$a_n = a_n^h + a_n^p$

a_n^h → homo

a_n^p → particular

$$a_n^h = \alpha_1(3^n)$$

$$a_n^p = -n - \frac{3}{2}$$

$$a_n = a_1(3^n) - n - \frac{3}{2} \qquad a_1 = 3$$

$$3 = a_1(3^1) - 1 - \frac{3}{2}$$

$$3 = 3a_1 - 1 - \frac{3}{2}$$

$$3 + \frac{3}{2} + 1 = 3a_1$$

$$a_1 = 11/6$$

$$\boxed{\frac{11}{6}(3^n) - n - \frac{3}{2}}$$



if lets say particular SOL was exponential

$$a_n = C r^n$$

and if it was trigonometric $a_n = C\cos(n) + D\sin(n)$